

Appendix S1 – Supplementary Information

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This document contains all the material necessary to support the main findings of the manuscript:

Beaver-engineering amplifies dissolved organic carbon concentration

Data collection and processing

Water residence time

We defined the water residence time (i.e., τ , seconds) using Equation 1:

$$\tau = \frac{V}{Q} \quad (1)$$

where Q is the stream discharge (m^3/s) and V is the water volume stored within the beaver's pond (m^3). We estimated stream water discharge using a semi-distributed hydrological catchment model (PREVAH; Precipitation-Runoff-Evapotranspiration HRU Model; Viviroli *et al.* 2009). We estimated pond water volume (V) by delineating the beaver pond boundary from aerial imagery and clipping the DEM to this polygon. With the QGIS's Volume Calculation tool we then quantified the volume enclosed within the pond footprint by integrating elevation differences across all DEM cells within the polygon. We validated this method by field surveys of six different beaver ponds.

Channel gradient

We estimated channel gradient between upstream and downstream locations by upstream and downstream elevations from the DEM using the “Extract values to point” tool. We then digitized polylines that follow the swisTLM3D (SwissTopo 2024) river line between each upstream-downstream pair and calculated their lengths (LL) with the QGIS field calculator. We computed slope as the elevation drop per horizontal distance using Equation 2:

$$channel\ gradient\ (\%) = \frac{(Upstream - Downstream_{altitude})}{LL} \cdot 100 \quad (2)$$

Solar radiation

We computed the Solar radiation ($\frac{Wh}{m^2}$) of the stream reaches with the Grass r.sun.insoltime algorithm using the *qgisprocess* package in R (Muenchow et al., 2017). This algorithm requires a digital surface model (DSM), a slope layer and an aspect layer. We calculated the DSM by combining a 2 m digital elevation model (DEM, SwissTopo 2022) and a 1m digital model of vegetation height (resampled to the resolution of the DEM) (FOEN, 2021). Slope and aspect were computed from the 2 m DEM using the *terra* package in R (Hijmans, 2025).

Primary producers variables

We sampled primary producers' abundance, namely phytoplankton, macrophyte and soil litter cover at 16 sites (Figure 1a, main manuscript). To estimate phytoplankton abundance, we

collected 50L of water from the main beaver pond and concentrated it to a 2 L sample using a 10 m mesh. In the laboratory, we counted and identified phytoplankton with a dual-magnification dark-field imaging microscope that captured all particles in the size range between ~ 10 m and ~ 1 cm using a 5.0E magnification objective. We estimated macrophyte abundance by counting each individual plant detected in the beaver ponds. We estimated soil litter percentage cover in a 1x5 m plot located at 0.5 m distance from the edge of the beaver pond in each site. We defined all dead plant material, such as leaves, but also twigs < 7 cm circumference, as soil litter.

Full Structural Causal Model specification

In this study we used the structural causal model (SCM) framework to answer the research questions (*sensu* Arif and MacNeil 2023). The SCM can be considered as a directed acyclic graph (DAG) with nodes and edges. The nodes indicate the relevant variables of the ecosystem and the edges indicate direct causal influences. The edges are acyclic because a cause is not at the same time an effect. We fitted the SCM with the Bayesian framework to accommodate the uncertainty inherent in ecological data and to incorporate prior distributions based on previous research and expert opinion (Gelman *et al.* 2013).

We specified the SCM model as a system of linear equations, where each equation represents the relationship between a response variable and its predictors. We statistically defined the full bayesian SCM as in Equation 1.

Given that the error in both the downstream and upstream DOC measurements was 0.2 mg C L^{-1} , we estimated the error (i.e., E) using Equation 3:

$$E \approx 0.283 \text{ mg/L} \quad (3)$$

Specifically, we treated the measurement error as missing data and imputed it using a normal distribution with a mean of 0 and a standard deviation of 0.283 mg/L. The overall model comprised a measurement model and an outcome multivariate model. First, we accounted for observational uncertainty by specifying a measurement model that links the observed $\Delta DOC_i^{obs/imputed}$ to a latent (true) DOC change, ΔDOC_i^{true} :

$$\textbf{Measurement model: } \Delta\text{DOC}_i^{\text{obs}} \sim \mathcal{N}(\Delta\text{DOC}_i, 0.283^2)$$

$$\textbf{Outcome multivariate model: } \Delta\text{DOC}_i \sim \text{Student-}t(\nu, \mu_i, \sigma_{\Delta\text{DOC}})$$

$$\begin{aligned} \mu_i = & \beta_1 \cdot \text{DOC}_{\text{upstream},i} \\ & + \beta_2 \cdot \text{Macrophytes}_i \\ & + \beta_3 \cdot \text{Phytoplankton}_i \\ & + \beta_4 \cdot \text{Litter Cover}_i \\ & + \beta_5 \cdot \text{WRT}_i \end{aligned}$$

$$\textbf{Predictors multivariate model:}$$

$$\begin{aligned} \text{Macrophytes}_i & \sim f(\text{solar radiation}_i, \text{WRT}_i, \dots) \\ \text{Phytoplankton}_i & \sim f(\text{Macrophytes}_i, \text{WRT}_i) \\ \text{Litter Cover}_i & \sim f(\text{solar radiation}_i) \\ \text{WRT}_i & \sim f(\text{DamHeight}_i, \text{NDams}_i, \dots) \\ & \vdots \end{aligned} \tag{2}$$

$$\begin{aligned} \textbf{Priors: } \alpha_{\text{summer}} & \sim \mathcal{N}(0, 10) \\ \alpha_{\text{winter}} & \sim \mathcal{N}(0, 7.5) \\ \beta_2, \beta_3, \beta_4, \beta_5 & \sim \mathcal{N}(0.05, 0.1) \\ \beta_6 & \sim \mathcal{N}(0, 0.2) \\ & \vdots \end{aligned}$$

$$\begin{aligned} \sigma_{\Delta\text{DOC}} & \sim \text{Student-}t(3, 0, 2.5) \\ \sigma_{\text{upstream}} & \sim \mathcal{N}(0, 0.5) \\ \nu & \sim \text{Gamma}(2, 0.1) \end{aligned}$$

(4)

We used the `mi()` function from the R package *brms* (version 2.22) to account for the measurement error of 0.283 mg/L on the response variable ΔDOC . We imputed missing data during model fitting by treating them as additional parameters. We ensured that missing data in both response and predictor variables were modeled rather than excluded by using the `mi()` function. This function treats missing values as additional parameters and estimates them jointly with the model.

We generated 20,000 (four chains run for 10,000 iterations discarding the first 3000 as burn-in) Markov chain Monte Carlo (MCMC) samples from the posterior distribution. We visually assessed if the SCM was a reasonably good description of the data by posterior predictive check and residuals (Figure S2 and Figure S3). All MCMC chains indicated convergence through falling within the threshold (i.e., $\hat{R} < 1.05$) specified by Gelman and Rubin (Gelman and Rubin 1992).

If posterior parameter distributions had more than 90% probability mass on either side of zero ($P(\beta > 0)$ or $P(\beta < 0)$), then we assumed that there was statistical evidence for the positive or negative effect of the predictor on the response variable.

We did all the data cleaning and statistical analysis within the R environment (version 4.5.1). We conducted data cleaning with the *tidyverse* (version 1.3.1) package and created figures using *ggplot2* (version 3.3.5). We used the R packages *brms* (version 2.16.1), *tidybayes* (version 3.0.2), and *bayesplot* (version 1.9.0) for Bayesian inference.

The beaver amplification factor (M): definition of floodplain area

We delineated the catchment area by automatically assigning pour points to stream outlets at each site and entering these, along with a D8 pointer raster based on a 25m DEM (Swiss Topo, 2004) into the *Watershed* algorithm in Whitebox (Lindsay, 2016) executed using the *Whitebox* package in R (Wu and Brown, 2022). We calculated the floodplain area of each site using a custom-made function in R based on the elevation differential of the local terrain and the nearest channel, adapted from a method developed by Clubb et al. (2017).

SCM: tested hypothesis

Table S1

Table S1. Tested hypothesis for the structural causal model (SCM) represented in Figure 1b, main document.

Causal path	Predictor		Response	Hypothesis
1	DOC upstream	up-	ΔDOC	If DOC upstream increases, then ΔDOC decreases because beaver ponds can act as sink, retaining or transforming DOC through processes such as microbial degradation, sedimentation, and photodegradation, thereby reducing downstream DOC concentrations relative to upstream inputs (Larson et al. 2007; Wohl 2013; Johnston 2014).
2	Macrophytes		ΔDOC	If macrophyte abundance increases, we expect ΔDOC to increase because the decomposition of macrophyte biomass releases labile dissolved organic carbon, enhancing downstream concentrations (Reitsema et al. 2018).
3	Phytoplankton		ΔDOC	If phytoplankton abundance increases, then ΔDOC increases is likely to increase, because phytoplankton contribute to DOC content through exudation during growth and release upon cell lysis, thereby enhancing downstream DOC levels relative to upstream inputs (Søndergaard et al. 1985; Sell and Overbeck 1992).
4	Soil cover	litter	ΔDOC	If soil litter cover increases, then ΔDOC increases as decomposition of litter releases dissolved organic compounds into the water (Aitkenhead et al. 1999; Cincotta et al. 2019).
5	Water residence time	resi-	ΔDOC	If water residence time increases, then ΔDOC increases, because longer water residence times allow for more extensive microbial decomposition and processing of dissolved organic carbon, enhancing downstream DOC concentrations (Jepsen et al. 2019; Abbasi et al. 2024).
6	Solar radiation	radia-	Phytoplankton	If solar radiation increases then phytoplankton abundance is expected to increase because greater light availability enhances photosynthetic rates (Reynolds 2006).
7	Solar radiation	radia-	Macrophytes	If solar radiation increases, then macrophyte abundance increases because higher light availability enhances photosynthetic activity and supports plant growth (Feuchtmayr et al. 2009).
8	Solar radiation	radia-	Litter cover	If solar radiation increases, as in summer, then soil litter cover decreases because lower rates of tree leaf fall coincide with enhanced decomposition driven by higher temperatures and solar radiation (Gessner et al. 1999).
9	Water residence time	resi-	Macrophytes	If water residence time increases, macrophyte abundance may increase

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Structural Causal Model: the priors

To ensure the model was not over-constrained by prior knowledge, we used weakly informative priors for all parameters. This approach allows the data to inform the model without assuming strong beliefs about the parameter values beforehand (Figure S1). We note, however, that for macrophytes, phytoplankton, and litter cover—variables measured in 16 of the 176 streams—we used more informative priors that incorporate established ecological knowledge about their influence on DOC concentrations (Figure S1 panel b).

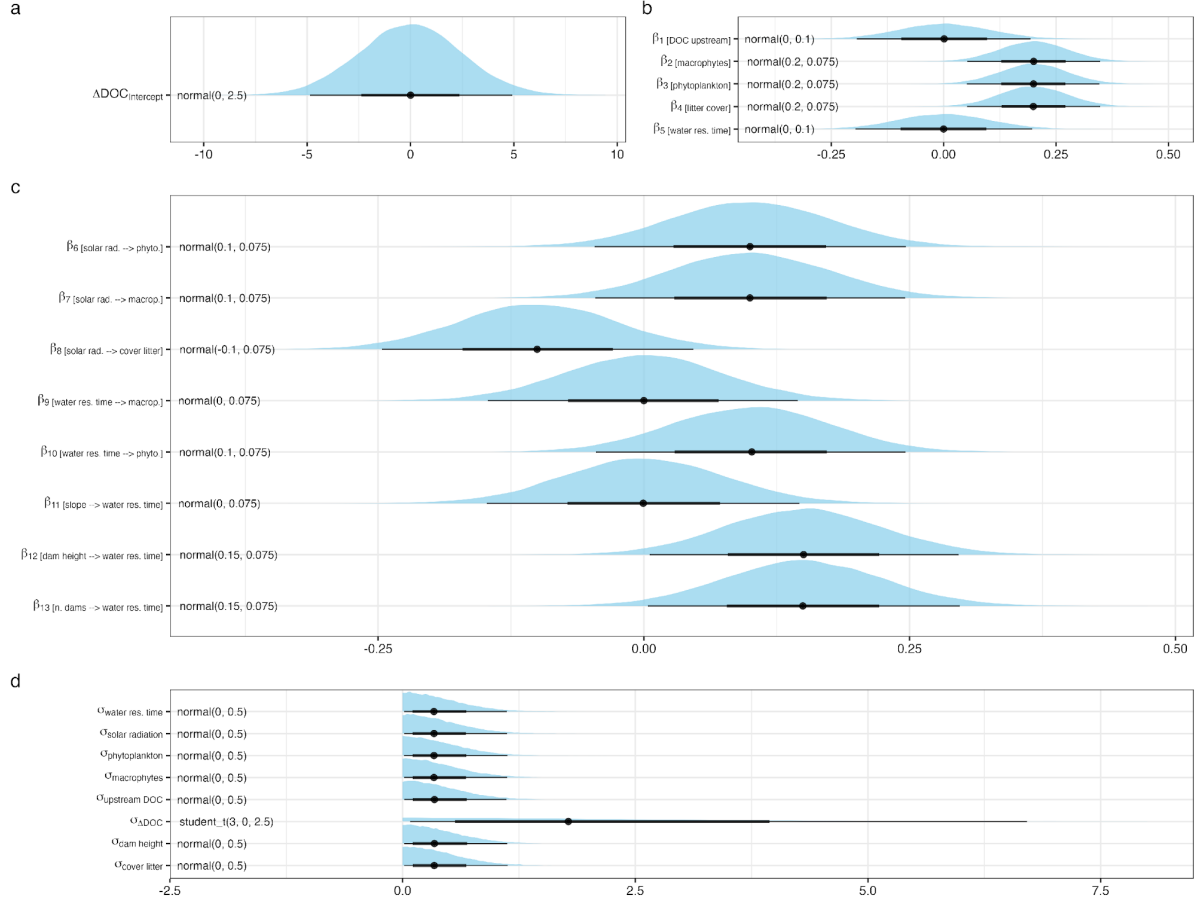


Figure S1: Half-eye plots display the prior distributions used in the model. The shaded areas (slabs) represent the probability density of each prior distribution, illustrating their shape and spread. The central dot indicates the median of the distribution, while the thick and thin horizontal lines represent the 66% and 95% credible intervals, respectively. Text labels to the left of each distribution indicate the corresponding prior specification.

Structural Causal Model: the residuals

We can evaluate the model fit by looking at the difference between observed data and their predictions. That are the residuals (r_i).

$$r_i = y_i - y_i^{\text{pred}}. \quad (5)$$

If the model is a reasonable representation of the observational process that originates the data then the residuals should look roughly randomly scattered in comparison to a horizontal line.

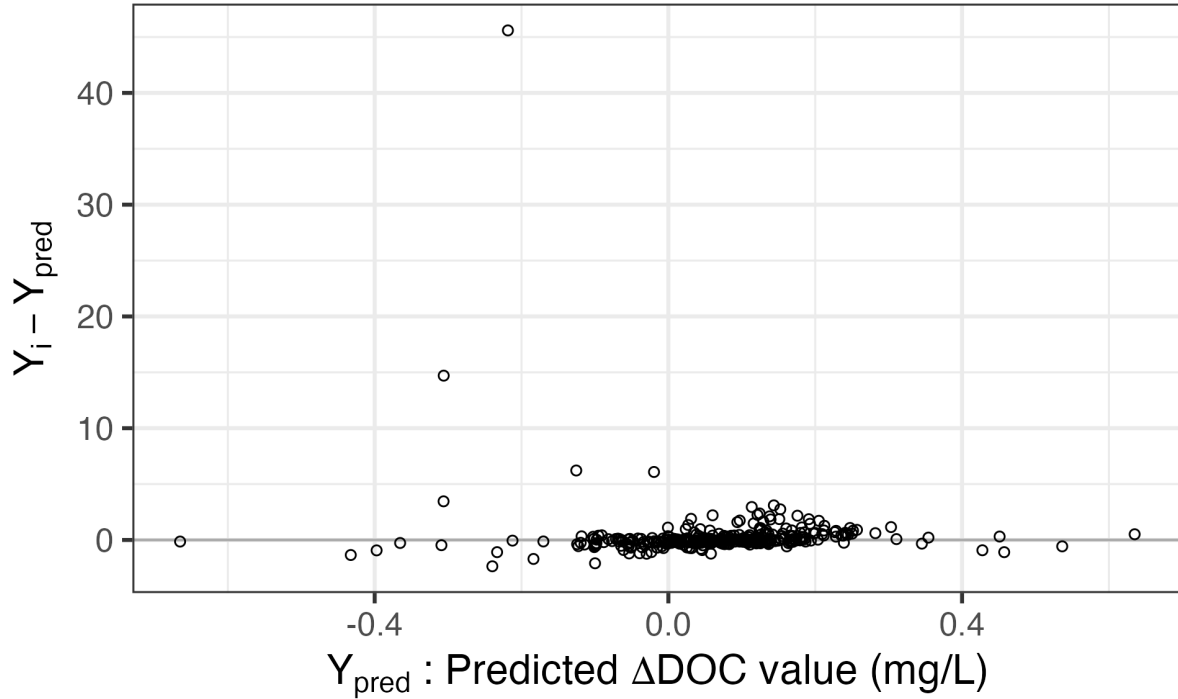


Figure S2: Residual plot showing the relationship between model-predicted ΔDOC values (mg L^{-1}) and corresponding residuals. Points represent the difference between observed and predicted values; a random scatter around zero (grey line) indicates good model fit and no systematic bias in predictions.

Structural Causal Model: posterior predictive checks

Posterior predictive checks assess how well the fitted SCM model reproduces observed data by comparing the observed values to data simulated from the model's posterior predictive distribution.

Structural Causal Model: posterior parameters distribution

References

- Abbasi M, Peacock M, Drakare S, et al. 2024. Water residence time is an important predictor of dissolved organic matter composition and drinking water treatability. *Water Research* **260**: 121910.
- Aitkenhead JA, Hope D, and Billett MF. 1999. The relationship between dissolved organic carbon in stream water and soil organic carbon pools at different spatial scales. *Hydrological Processes* **13**: 1289–302.
- Arif S and MacNeil MA. 2023. Applying the structural causal model framework for observational causal inference in ecology. *Ecological Monographs* **93**: e1554.
- Cincotta MM, Perdrial JN, Shavitz A, et al. 2019. Soil aggregates as a source of dissolved organic carbon to streams: An experimental study on the effect of solution chemistry on water extractable carbon. *Frontiers in Environmental Science* **7**.
- Clubb, F., Mudd, S., Milodowski, D., Valters, D. A., Slater, L. J., Hurst, M. D., & Limaye,

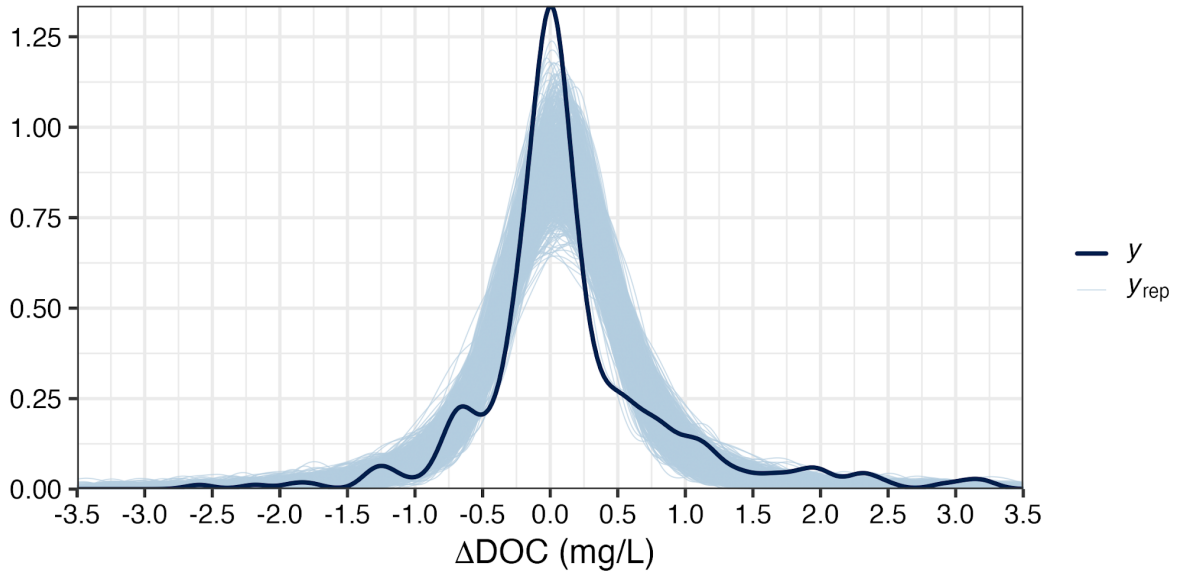


Figure S3: Posterior predictive checks of the mean and standard deviation statistics for the response variable delta DOC, with 1000 posterior draws used to generate the predictive distribution. This plot provides a visual assessment of the model's fit by comparing the observed data (dark blue) against the model's predictions (light blue) from the posterior distribution.

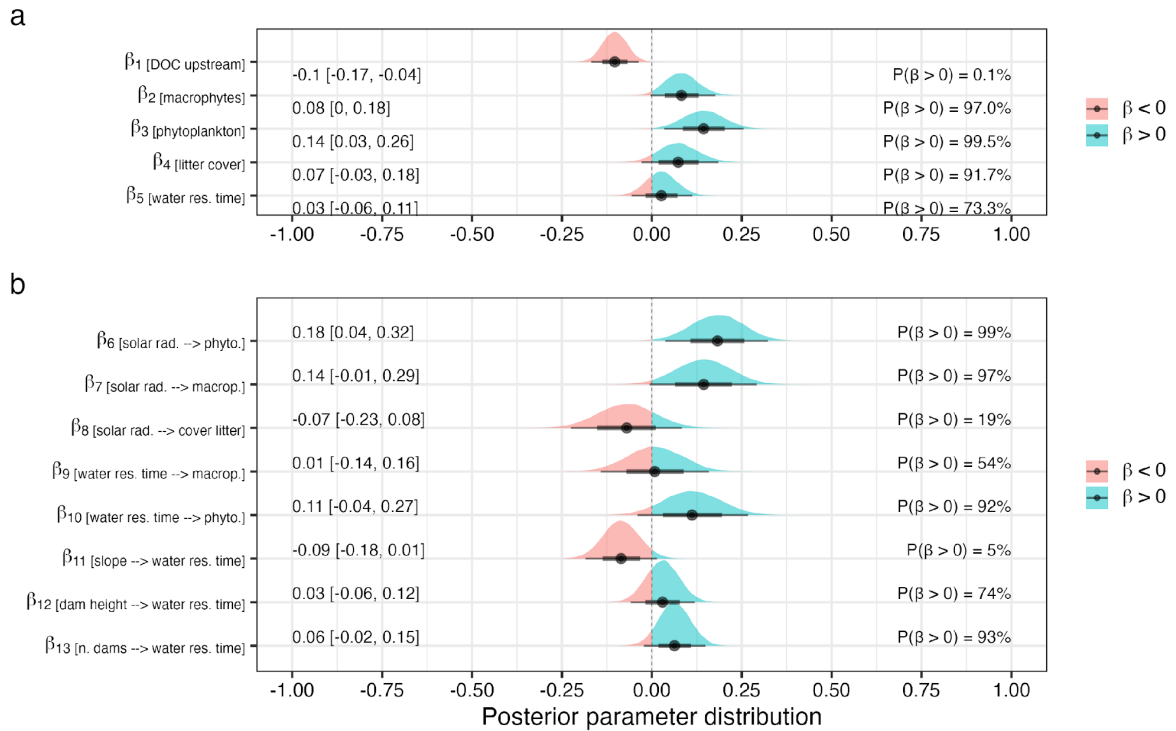


Figure S4: Posterior parameters distributions of the fitted SCM. Median shown with point; 66% and 95% credible intervals shown with black bars. Parameters that have more than 90% probability mass on either side of zero are considered to have a positive or negative influence on the response variable. a) The direct effects on ΔDOC b) The indirect effects on ΔDOC .

- A. B. 2017. Geomorphometric delineation of floodplains and terraces from objectively defined topographic thresholds. *Earth Surface Dynamics*, 5(3), 369-385.
- Dittbrenner BJ, Schilling JW, Torgersen CE, and Lawler JJ. 2022. Relocated beaver can increase water storage and decrease stream temperature in headwater streams. *Ecosphere* **13**: e4168.
- Feuchtmayr H, Moran R, Hatton K, et al. 2009. Global warming and eutrophication: effects on water chemistry and autotrophic communities in experimental hypertrophic shallow lake mesocosms. *Journal of Applied Ecology* **46**: 713–23.
- Finger D, Schmid M, and Wüest A. 2007. Comparing effects of oligotrophication and upstream hydropower dams on plankton and productivity in perialpine lakes. *Water Resources Research* **43**.
- FOEN, 2021. Vegetation height model NFI.
- Gelman A, Carlin JB, Stern HS, et al. 2013. *Bayesian Data Analysis*. CRC Press.
- Gelman A and Rubin DB. 1992. Inference from iterative simulation using multiple sequences. *Statistical Science* **7**: 457–72.
- Gessner MO, Chauvet E, and Dobson M. 1999. A perspective on leaf litter breakdown in streams. *Oikos* **85**: 377–84.
- Hijmans, R. (2025). terra: Spatial Data Analysis. R package version 1.8-87.
- Jepsen SM, Harmon TC, Sadro S, et al. 2019. Water residence time (age) and flow path exert synchronous effects on annual characteristics of dissolved organic carbon in terrestrial runoff. *Science of The Total Environment* **656**: 1223–37.
- Johnston CA. 2014. Beaver pond effects on carbon storage in soils. *Geoderma* **213**: 371–8.
- Larson JH, Frost PC, Zheng Z, et al. 2007. Effects of upstream lakes on dissolved organic matter in streams. *Limnology and Oceanography* **52**: 60–9.
- Lindsay, J. B. 2016. Whitebox GAT: A case study in geomorphometric analysis. *Computers & Geosciences*, 95, 75-84.
- Majerova M, Neilson BT, Schmadel NM, et al. 2015. Impacts of beaver dams on hydrologic and temperature regimes in a mountain stream. *Hydrology and Earth System Sciences* **19**: 3541–56.
- Manolaki P, Olesen A, Hvidt BG, et al. 2021. Investigating emergent macrophytes establishment rate and propagation towards constructed wetlands efficacy optimization. *Knowledge & Management of Aquatic Ecosystems* 23.
- McGuire KJ, McDonnell JJ, Weiler M, et al. 2005. The role of topography on catchment-scale water residence time. *Water Resources Research* **41**.
- Nyssen J, Pontzele J, and Billi P. 2011. Effect of beaver dams on the hydrology of small mountain streams: Example from the chevral in the ourthe orientale basin, ardennes, belgium. *Journal of Hydrology* **402**: 92–102.
- Reitsema RE, Meire P, and Schoelynck J. 2018. The future of freshwater macrophytes in a changing world: Dissolved organic carbon quantity and quality and its interactions with macrophytes. *Frontiers in Plant Science* **9**.
- Reynolds CS. 2006. *The ecology of phytoplankton*. Cambridge: Cambridge University Press.

- Sell AF and Overbeck J. 1992. Exudates: Phytoplankton-bacterioplankton interactions in plusee. *Journal of Plankton Research* **14**: 1199–215.
- Søndergaard M, Riemann B, and Jørgensen NOG. 1985. Extracellular organic carbon (EOC) released by phytoplankton and bacterial production. *Oikos* **45**: 323–32.
- SwissTopo. 2022. swissALTI3D.
- Vats D and Knudson C. 2021. Revisiting the gelmanrubin diagnostic. *Statistical Science* **36**: 518–29.
- Viviroli D, Zappa M, Gurtz J, and Weingartner R. 2009. An introduction to the hydrological modelling system PREVAH and its pre- and post-processing-tools. *Environmental Modelling & Software* **24**: 1209–22.
- Westbrook CJ, Cooper DJ, and Baker BW. 2006. Beaver dams and overbank floods influence groundwatersurface water interactions of a Rocky Mountain riparian area. *Water Resources Research* **42**.
- Wohl E. 2013. Landscape-scale carbon storage associated with beaver dams. *Geophysical Research Letters* **40**: 3631–6.
- Wu, Q., Brown, A. 2022. 'whitebox': 'WhiteboxTools' R Frontend. R package, version 2.2.0.